

The Utility Problem

Utilities are in various phases of smart meter implementation. If smart meters are not fully implemented, it is difficult to obtain a realistic load profile of energy consumption. If such a profile is required and consumption readings are taken once every six months or once a year, it is necessary to work with load profile estimation, also known as standard consumption pattern. These load profile estimates (consumption patterns) have a wide range of uses. Such uses are, for example, uses in analysis for periods shorter than the meter reading period; these include analyses of grid load estimates for different time intervals, energy balances, load flows, etc.

Several methods of estimating these values can be used. These methods range from simpler ones that take, for example, the average value of the measured data to more complex ones that use complex mathematical operations such as probability densities, but also many others. For such calculations, it is necessary to have at least some sample of real/measured time-series data on which to base the calculations. The larger this sample is (i.e. more meters with different types of consumption, with or without electricity heating, with or without electric hot water heating), but also the longer the time period (time period over which this data is measured), the more accurate the output of the calculation itself is expected to be.

The Data Solutions and Methods

By application of the described methods, analysts would get 15-minute or hourly demand allocations for a group of customers. These allocations can then be applied to annual or semi-annual consumption amounts in order to get a more granular/estimated consumption shape.

The following description aims for two methods:

1. Typical Loads Diagrams (TLD):

This method is used to create load profiles for selected groups of customers with comparable electricity consumption patterns. The customers are divided into several classes, which are determined on the basis of the type of consumption (e.g. for households with similar appliances - water heater, households with or without electricity heating, residential vs. commercial consumers, etc.). The output for a given class is 8760 values (= 365 days * 24 hours) between 0 (no consumption) and 1 (maximum consumption), determining the shape of the consumption pattern.

The compilation is done from type samples from the same class - real/measured time-series data are installed at specimen consumers. Based on these values, typical load diagrams are calculated for each selected class. First, a so-called Normalized Typical Load Diagram (TLDn) is created. This is the 8760 relative values of the average hourly consumption in a year relative to the value of the annual maximum of the average hourly consumption determined from the sample measurements.

The next step in constructing the TLD of a customer is to take into account its annual consumption and the outdoor temperature trend. With this approach - an estimated 15-min or hourly consumption curve can be created for any consumer with annual or semi-annual consumption reading. Such disaggregated values can then be used for further analytics use cases.

2. Probabilistic approach:

Another method to estimate customer load is to use probability densities. Again, real/metered time-series data are needed. The customer sites are divided into classes according to tariffs and circuit breakers. The measurements of all customer sites of a given class for a monthly period are taken and the data is grouped by time regardless of the day - since the quarter hour is used as the smallest time unit, this gives $4 * 24 = 96$ time groups. For each such group, the probability distribution (empirical density) of the power is determined. This results in 96 probability densities - each for one quarter-hour window.

To apply these probabilities to multiple customers for a particular feeder, for example, the following procedure is used. The first customer is taken and its category is determined and probability densities corresponding to that category are assigned. The next step is to determine these probabilities for the two customers together - i.e. looking for the distribution of the sum of the individual customer that is assumed to be independent. The summation probability density is a convolution of the summation densities. For multiple customers, a convolution is performed each time another customer is "added" to the convolution already computed (the probability densities for the whole feeder might be obtained step by step). Any statistic on the distribution given by the resulting probability densities can be chosen as the output, e.g. mean, maximum or other percentiles - most commonly, we use the 98th percentile, which we interpret as an estimate of the load that will be exceeded only 2% of the time. At the same time, this reduces the effect of erroneous outliers on the output values. The time course of this statistic within a day is constructed from 96 consecutive time window values.

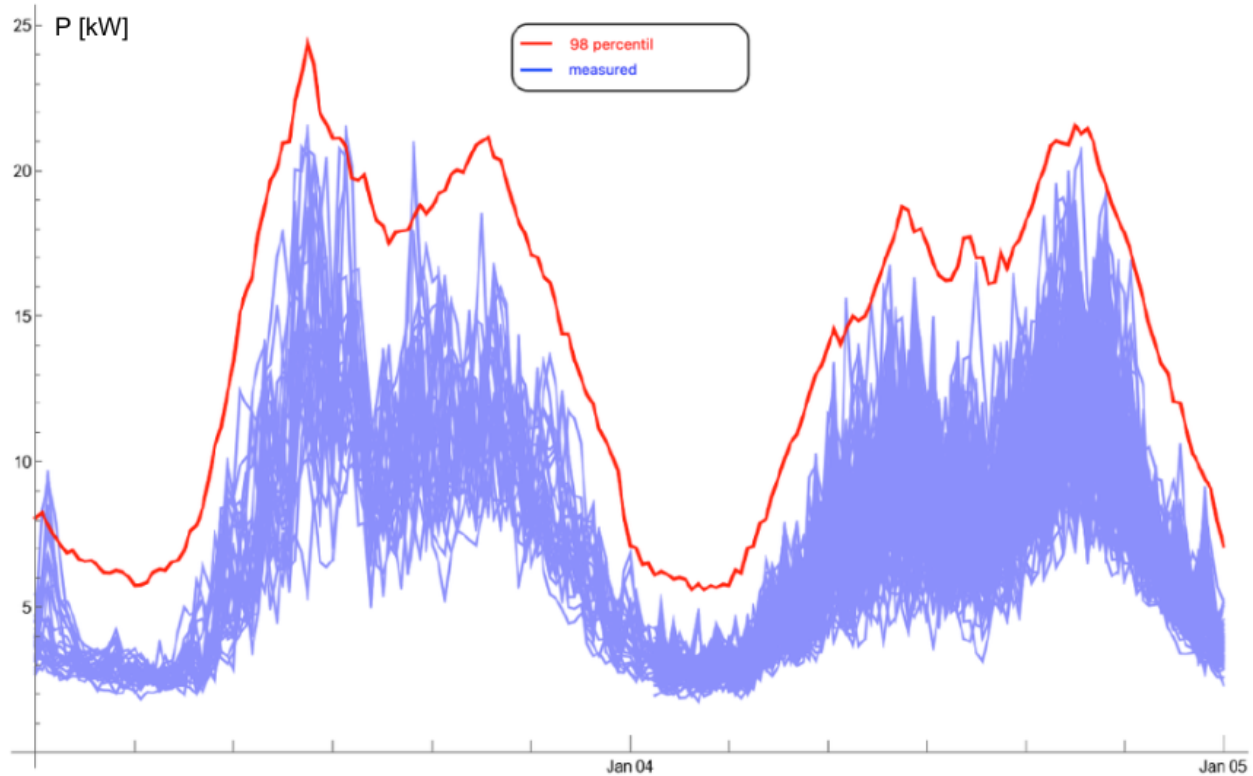


Figure 1: Measured data and the final 98th percentile

For a more accurate estimation, it is advisable to split the measurement further. Thus, it is necessary to calculate probability functions separately for working and non-working days, appropriately clustering different types of consumers (but in order to have enough measured data) and circuit breakers. It is also appropriate to take into account the outdoor temperature.

The Awesense Energy Transition Platform enables these types of analyses. Thanks to the integration and synchronization of a wide range of data, all the necessary data is easily available and can be worked with straight away without the need to deal with data cleaning. It is also possible to use pre-built functions that simplify such calculations.

What are the data needed for this analysis:

1. Timeseries data for at least part of the customers.
The data is very easy to obtain for selected customers - using the timeseries data calling function. It is also possible to use the predefined data aggregation function, which converts, for example, 15 minutes of data into hourly data.
2. Additional static data of the customers - for example, circuit breaker size.
This data is also part of the Awesense data model and is therefore very easy to obtain.

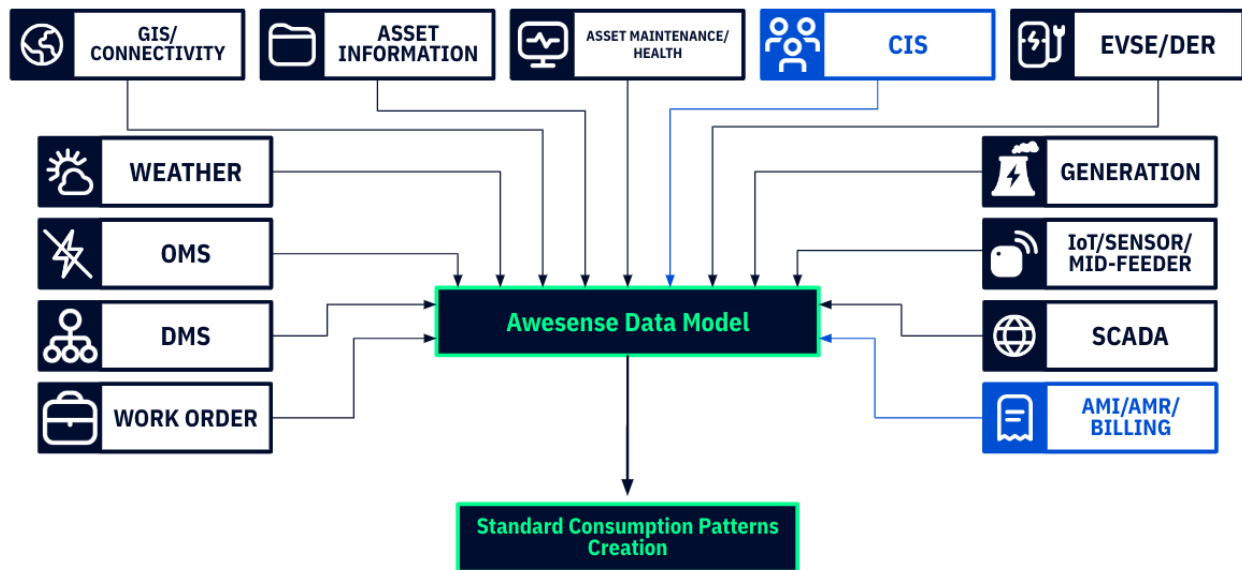
Standard Consumption Patterns Creation

[UC35-01] - Awesense Energy Transition Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.